PDE-Constrained Optimization on an External Mesh: Theory, Algorithms, and Experiments

Fachseminar Optimization — Legacy FEniCS Implementation

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Abstract

We study a linear–quadratic elliptic PDE-constrained optimization problem on $\Omega = (0,1)^2$ with a circular control region $\omega \subset \Omega$. We develop the variational formulation, discrete operators, gradient and adjoint equations, and discuss smooth first-order methods: fixed-step gradient descent using a Lipschitz estimate, Barzilai–Borwein (BB), and Nesterov acceleration with restart. We use an external Gmsh mesh with exact circular subdomain tags and present numerical results, sensitivity to β, and mesh-independence. All figures and results are reproducible with the provided Python scripts (legacy FEniCS).

1 Problem statement and variational formulation

Let $\Omega = (0,1)^2$, $\omega \subset \Omega$ a disk of radius $r = 0.1$, and $y_d \in L^2(\Omega)$. Given $\beta = 0.001 > 0$, we minimize

$$\min_{u \in L^2(\omega)} J(y,u) := \tfrac{1}{2}\|y - y_d\|^2_{L^2(\Omega)} + \tfrac{\beta}{2}\|u\|^2_{L^2(\omega)} \quad \text{s.t.} \quad -\Delta y = \chi_\omega\, u \text{ in } \Omega,\ y = 0 \text{ on } \partial\Omega. \tag{1}$$

In weak form: find $y \in V := H^1_0(\Omega)$ s.t. $\int_\Omega \nabla y \cdot \nabla v\, dx = \int_\omega uv\, dx$ for all $v \in V$.

Adjoint and reduced derivatives. For the adjoint $p \in V$ solving $\int_\Omega \nabla v \cdot \nabla p\, dx = \int_\Omega (y - y_d) v\, dx$, the reduced gradient and Hessian in $U = L^2(\omega)$ are

$$\nabla f(u) = \Pi_\omega(p) + \beta u, \tag{2}$$

$$Hv = \Pi_\omega\big(A^{-1} M A^{-1} B v\big) + \beta v, \tag{3}$$

where A is the stiffness on V, M the mass on V, B the control coupling, and Π_ω the L^2-projection/restriction to ω.

2 Discretization and external mesh

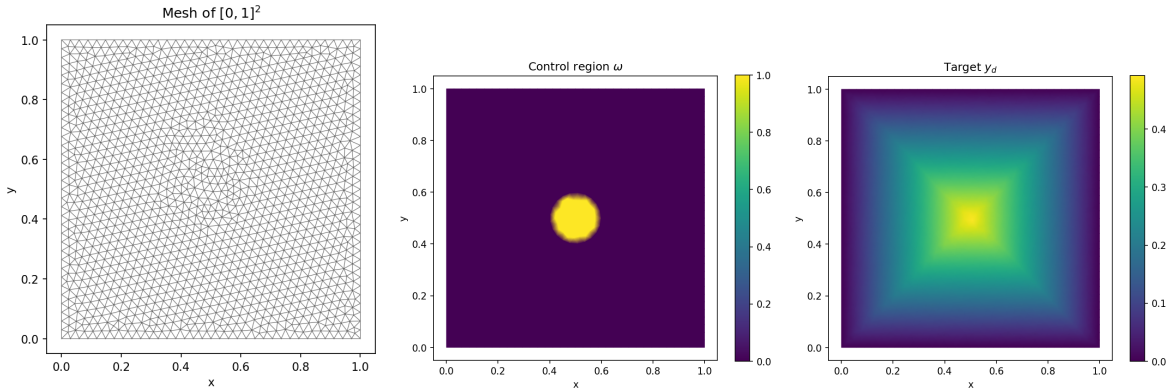
We use $\mathbb{P}_1$ elements on a Gmsh triangulation with exact circular tags. Import via XDMF; assemble B and the control L^2 inner product on $dx(\omega)$ only; A and M on dx.

3 First-order methods

We minimize f in the M_ω-inner product. We compare: fixed-step GD with $\alpha = 1/L$, BB (alternating BB1/BB2), and Nesterov with restart.

3.1 Gradient descent with fixed step

Given $L \geq \lambda_{\max}(H)$, update $u^{k+1} = u^k - \frac{1}{L}\nabla f(u^k)$. For μ-strongly convex f, $f(u^k) - f^* \leq (1 - \mu/L)^k (f(u^0) - f^*)$.



(a) Mesh (b) Control region ω (c) Target y_d

Figure 1: External mesh and problem data. Triangles: 2730, $h \approx 0.027$.

Algorithm 1 Fixed-step GD

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Estimate L by power iteration on H  set α = 1/L
for k = 0, 1, ... do
    Solve A y^k = B u^k; A p^k = M(y^k − y_d)
    g^k ← M_ω^{-1} B^T p^k + β u^k; u^{k+1} ← u^k − α g^k
    if ‖g^k‖/‖g^0‖ < ε then break
    end if
end for
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3.2 Barzilai–Borwein

With $s^{k-1} = u^k - u^{k-1}$, $y^{k-1} = g^k - g^{k-1}$, use BB1/BB2 step sizes and $u^{k+1} = u^k - \alpha_k g^k$. See [1].

3.3 Nesterov acceleration

FISTA-style extrapolation with optional restart when f increases; see [3, 2].

4 Implementation notes

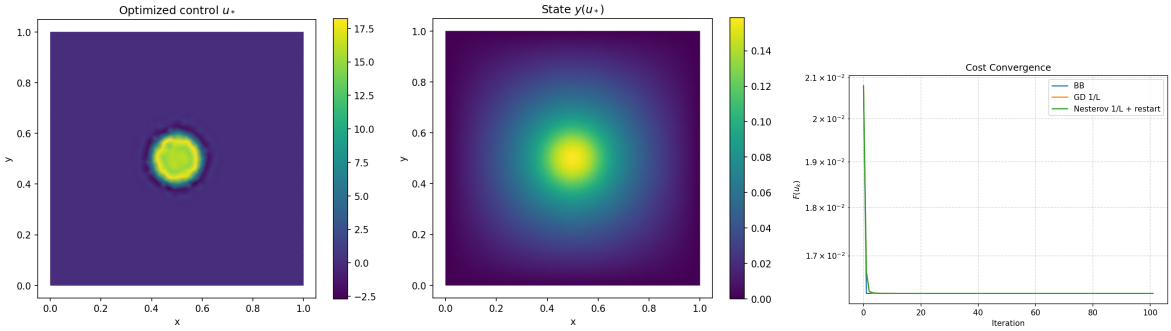
Assembly of A, M, B, M_ω; reuse sparse factorizations of A and M_ω; power iteration for L. Stopping by gradient norm.

5 Numerical results

Unless stated: $\beta = 0.001$, $r = 0.1$. We report optimized control and state, and convergence.

6 Sensitivity and mesh-independence (optional)

Vary β and r; observe stronger controls for small β and slower GD unless L adapts. Refinements confirm stable iteration counts for BB/Nesterov.



(a) u^* (b) $y(u^*)$ (c) $F(u^k)$

Figure 2: Optimized control/state and objective history.

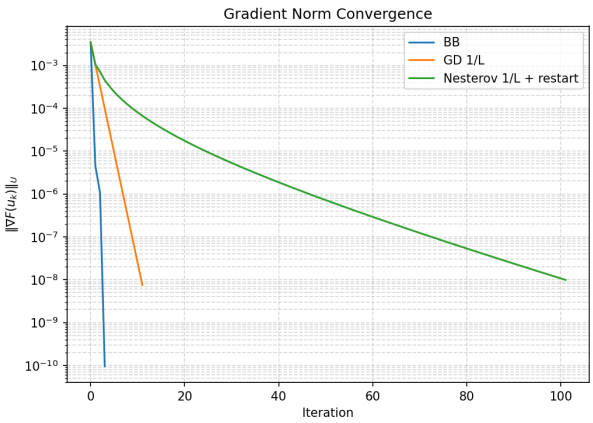


Figure 3: Gradient norm vs iterations (log scale).

7 Conclusions

External meshing with exact ω improves accuracy near the interface. BB often offers the best time/iteration trade-off; Nesterov is robust with restarts; fixed-step GD is dependable with a good L.

References

[1] J. Barzilai and J. M. Borwein, *Two-point step size gradient methods*, IMA Journal of Numerical Analysis, 8 (1988), pp. 141–148.

[2] A. Beck and M. Teboulle, *A fast iterative shrinkage-thresholding algorithm for linear inverse problems*, SIAM Journal on Imaging Sciences, 2 (2009), pp. 183–202.

[3] Y. Nesterov, *Introductory Lectures on Convex Optimization*, Springer, 2004.